Spatial and Demographic Equity in Boston's Bluebikes Network

Midterm Written Report

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Research Question

Our research question is: Across the City of Boston in 2024, do Bluebikes stations and trips concentrate in higher-income neighborhoods, and who lives in the areas with the least bikeshare activity? This question gets at all the ideas we were interested in when we proposed this topic, but puts some specificity to a lot of ambiguous terms. In particular, we were initially focused on *access* to Bluebikes. Now, we've defined two separate metrics to measure access, station location and trip volume. Station location helps us understand how convenient trips are for different groups of people, while trip volume helps us understand the patterns that are emerging as a result of station location and Bluebike demand in a given area.

Data

We draw on two sources. The first is Bluebikes' monthly trip data. We use the 12 files for 2024, which record each trip's start and stop time, start and stop station, vehicle type, and whether the rider is a member. The same page provides station data: name, latitude and longitude, municipality, seasonality, dock count, and station ID. The second source is the American Community Survey (ACS) 2020–2024 5-year estimates of income for Boston census tracts. Later, we plan to add more granular data published by the City of Boston.

To clean the data, we removed 22,733 of the 4,751,790 trips (0.48%) using four rules. First, we dropped 1,435 trips (0.03%) with no start station, since station volume is measured in outbound trips and these cannot be assigned to a station. As a result, we kept trips with no end station. Second, we dropped 17,473 trips (0.37% of the remaining trips) lasting under one minute, which were likely accidental unlocks rather than genuine use. Third, we dropped 3,702 trips (0.08% of the remaining trips) lasting over 24 hours, which were likely bikes not returned

by the original rider. Finally, we removed 123 duplicates (less than 0.01%) that appeared in two month files because they began in one month and ended in the next. No trips began in 2023 and continued into 2024.

Methods

After cleaning, we count trips beginning at each station per month as a measure of station use. We then download the ACS 2020–2024 5-year estimates for Suffolk County census tracts, project them to EPSG:2249 (Massachusetts Mainland, in feet), and keep tracts whose centroids fall within Boston. After projecting the station locations to the same CRS, we spatially join each station to its tract. This lets us sum trips per tract and compute trips per 1,000 residents. It also allows us to plot a station's trip volume against the median income in its tract.

We use these projected data to map station locations, map median income by tract with stations overlaid, and map a heatmap of 2024 station usage. We also chart the straight-line distance between start and end stations, and time-of-day histograms for weekdays and weekends, with member and casual riders overlaid.

Early Results

Two findings stand out. First, stations cluster in higher-income tracts, shown as the darker red and orange areas on our map. Confirming this, our plot of station usage against median tract income shows a grouping of stations along the top-left side of the graph. This side corresponds to stations where the median income is high but the total usage is relatively low. This is important because it shows that even when the demand for a given station isn't necessarily demonstrated, stations are still concentrated in high-income areas. Using Bluebikes requires both a station near

the start and end of a trip and the ability to afford a ride, so station density is a key measure of geographic access. Second, usage follows clear patterns that sharpen this concern. The vast majority of trips are under 1.5 miles, so in areas with sparse stations the most common use of Bluebikes may not be possible. Weekday usage also spikes just before 9 a.m. and after 5 p.m., suggesting commuting, which may be less practical where workplaces are far from stations. Stations are densest in the central business district, where a typical worker may be a higher-income office worker.

Plan to Finish

First, we will deepen the link between demographic and Bluebikes data. Using our existing spatial join, we will examine the economic, racial, and age profiles of communities around stations and when usage peaks in different areas. We will also replace tract-level ACS data with the City of Boston's more granular dataset. Second, we will include the municipalities surrounding Boston, which we currently exclude to simplify the midterm analysis. Third, we will conduct a brief literature review of methods used in micromobility research and compare low-usage outlying areas to the general Boston population.